srh Berlin University of Applied Sciences

Master Thesis Defense

Bridging the Gap between ROS 2 and Additive Manufacturing: A Real-Time Control Approach for Continuous Deposition Tasks

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Presentation Outline

- Introduction & Motivation
- Research Gap
- Objectives & Research Questions
- Literature Insights
- System Architecture
- Methodology
- Experiments & Protocols
- Results (Naïve vs Windowed)
- Cost Analysis
- Discussion
- Limitations & Future Work
- Conclusion

Introduction

Introduction

- Additive Manufacturing (AM) is a key technology in Industry 4.0, enabling flexible and customized production.
- Using robotic arms for AM allows larger parts, curved surfaces, and reduced support structures.
- The main challenge is keeping robot motion and extrusion synchronized in real time.
- Traditional G-code streaming often causes delays, starvation, and bead defects.
- ROS 2, with its real-time communication and modular design, is a strong solution.
- This thesis develops a ROS 2 coordinator node with an ack-aware windowing strategy to achieve continuous, starvation-free deposition on a low-cost robotic setup.

Motivation: Building a Needed Capability for Our Lab

- Our university lab had an **igus ReBeL robotic arm** but lacked a multi-axis 3D printing platform for research.
- The goal was to create a functional system using the resources we had. This involved integrating the existing robot with affordable, widely available components like an **E3D extruder** and a standard **SKR controller** board running Marlin firmware.
- This practical integration effort immediately exposed a deep technical challenge: the fundamental mismatch between the robot's real-time motion streaming and the extruder's buffered G-code planner.
- This project is therefore not just about building a printer; it is about **solving the critical synchronization problem** that arises when you bridge industrial robotics with open-source additive manufacturing hardware.

Objectives & Research Questions

Objectives:

- Build ROS 2 coordinator node at 50 Hz.
- Synchronize robot motion (CRI) & extrusion (Marlin).
- Implement ack-aware in-flight window (8–12 lines).
- Provide reproducible pipeline (configs, logs, repo).

Research Questions:

1. What synchronization settings enable continuous deposition?

2. What path error is achieved at TCP $\leq$ 0.2 m/s?

3. How robust is the pipeline against serial & thermal faults?

Literature Review

Literature Review

1. Robotic Additive Manufacturing (AM)

- Extends FFF beyond Cartesian printers However, prior works rely on custom toolchains and are not reproducible with standard slicer G-code, multi-axis deposition, larger build volume, reduced supports (Ding 2016; Pan 2019). Used in WAAM for aerospace/industrial parts.
- Gap: prior works rely on custom toolchains, not reproducible with standard slicer G-code.

2. Synchronization Challenges

- Bead quality depends on motion–extrusion synchronization.
- Problems: starvation (planner empty), overflow (queue bloat). Timing drift directly linked to bead width & bonding (Coogan & Kazmer, 2019). No reproducible quantification of timing bead continuity link.

3. Middleware & Firmware

- ROS 2: DDS-based middleware, QoS, modular integration → promising for AM.
- Prior ROS 2 studies: focus on motion planning / WAAM, not extrusion-level synchronization.
- Marlin firmware: buffered G-code with ack ("ok"), risk of starvation/overflow.

4. Gap in Literature

- No procedures for ack-aware, windowed streaming in robotic FFF. Lack of open, reproducible benchmarks linking timing metrics to bead quality.
- This thesis: bridges ROS 2 + Marlin with ack-aware windowing, proves starvation-free deposition, and contributes open logs/configs.

Research Gap

What Exists in Literature	What is Missing	My Thesis Contribution
Multi-axis FFF Demos	A standard, reproducible pipeline from slicer G-code to synchronized robot motion.	End-to-End Open Pipeline: G-code → synchronized ROS 2 streams (CRI + Marlin).
ROS 2 QoS/Latency Studies	Application of timing studies to extrusion controller synchronization.	Ack-Aware Synchronization Protocol: A bounded in-flight window policy to mitigate USB jitter and prevent planner starvation.
Proprietary/WAAM Systems	Open-source, reproducible baselines for polymer FFF on low-cost hardware.	Full Hardware/Software Stack: Public GitHub repo with all configs, logs, and calibration data for the igus ReBeL + SKR Mini E3 + Hemera XS system.

Methodology

Methodology: The 50 Hz Synchronization Loop

1

Interpret

Parse G-code segment and compute Instantaneous TCP speed v(t_k).

2

Calculate:

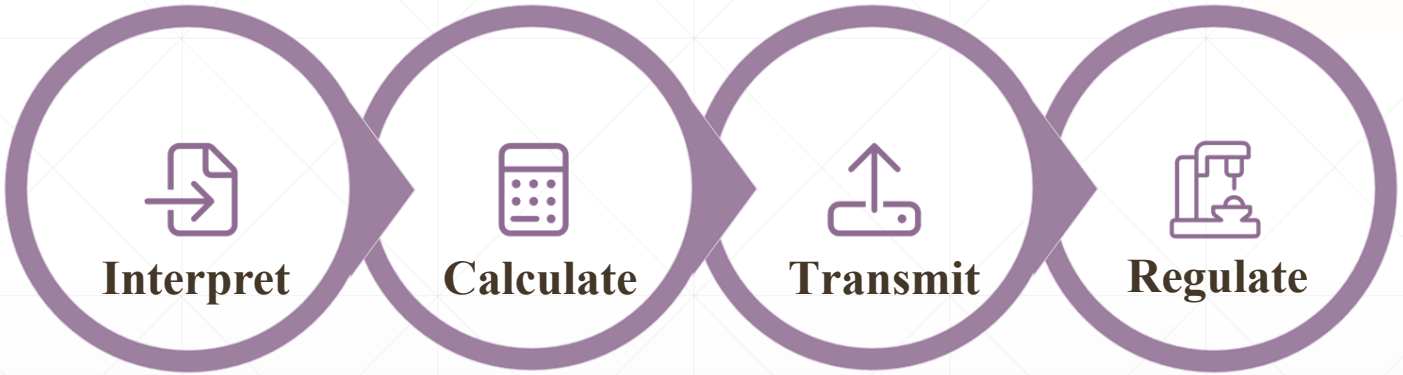
Compute the required extrusion increment.

3

Transmit:

Send synchronized commands:

- **Robot:** Cartesian pose via CRI interface.
- **Extruder:** G1 E<ΔE> F<ΔF> (relative extrusion) via USB.



4

Wait:

Listen for Marlin's "**ok**" acknowledgment.

5

Regulate:

Enforce the **bounded in-flight window** [8, 12].

- **If $N < N_{min}$:** Send a burst of commands to refill the planner.
- **If $N >= N_{max}$:** Throttle sending until acknowledgments are received.

⚙️ Methodology Ack-Aware Windowing Protocol

1

Control Loop (every 20 ms):

- Sample robot pose via CRI.
- Compute ΔE using extrusion model.
- Send motion + extrusion commands.
- Await "ok" acknowledgment from Marlin.

2

In-Flight Window Strategy:

- Maintain 8–12 unacknowledged commands.
- **Backlog Prevention:** Stop sending if $N \geq 12$.
- **Starvation Prevention:** Burst send if $N \leq 8$.
- Absorbs OS/USB jitter for smooth command flow.

Theoretical Framework: From Speed to Extrusion

Volumetric Flow Rate (Q):

$$Q(t) = w \cdot h \cdot v(t) \;\; [mm^3/s]$$

Where:

- w= bead width
- h= layer height
- $v(t)$= robot tool speed

Filament Feed Rate ($\dot{E}$):

$$\dot{E}(t) = \frac{Q(t)}{A_f} \; [mm/s] \times \; \Delta t$$

Where:

- A_f is filament cross-setional area

- It links **motion speed** to the **volume of polymer that must be deposited** to maintain a correct bead cross-section.
- This was the base equation to compute how much filament must be extruded per mm of nozzle motion.

- It converts **required plastic flow** into **filament feed motion**.
- With filament diameter known (1.75 mm), I computed how fast the extruder motor must push the filament.

Calibration & Safety: Ensuring a Valid Baseline

Calibration (Prerequisite for Valid Results):

- Thermal Stability: PID autotuning (M303, M301) at 240°C for PETG.
- Volumetric Accuracy: E-steps calibrated via 100mm filament feed test.
- Process Window: Locked to PETG, nozzle 0.4mm, w=0.48mm, h=0.20mm.

PID: Kp=19.10, Ki = 1.02, Kd = 85

E-steps: 397 steps/mm

Safety Implementation

Firmware Guards:

- M302 S220 (Blocks cold extrusion)
- EMERGENCY_PARSER (Ensures M112 is always recognized)
- Thermal Runaway Protection (Marlin firmware)

Host Watchdogs:

- Serial timeout (10s without "ok" triggers pause)
- Temperature deviation monitor

Hardware Safety:

- External E-stop button cutting power to the robot controller.
- Strain relief on cables to prevent snagging.

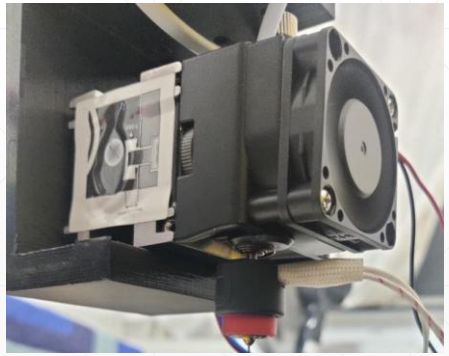
System Design

System Hardware Architecture

Hardware Integration:

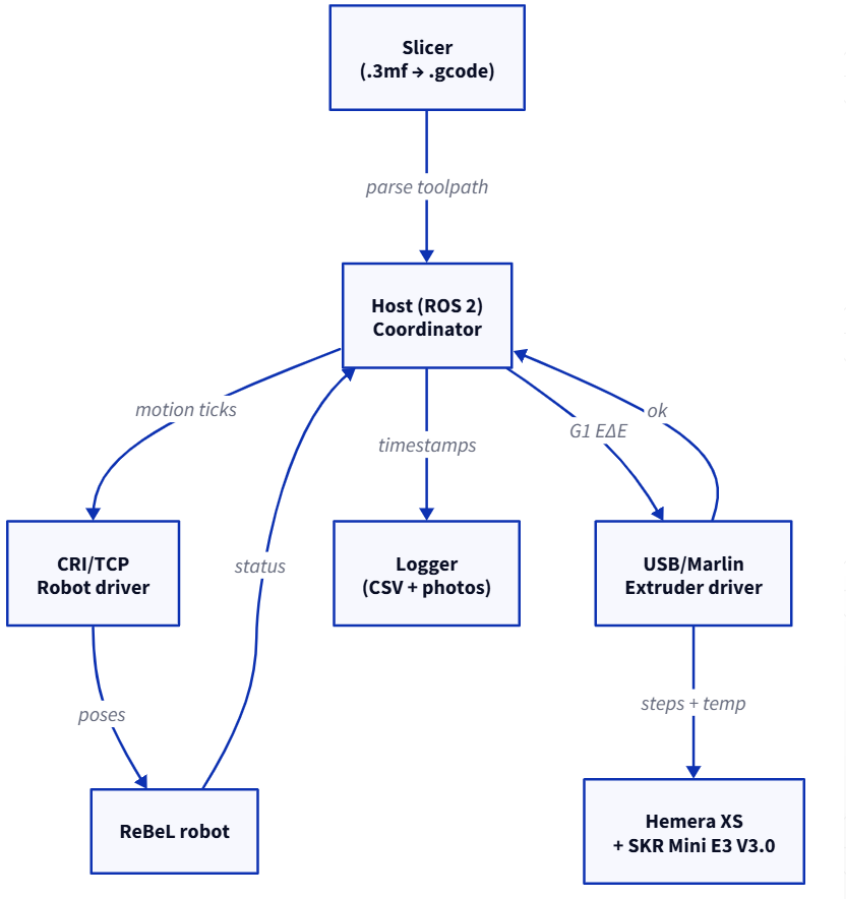
- **Robot:** igus ReBeL 6-DOF arm (TCP ≤ 0.2 m/s)
- **Extruder:** E3D Revo Hemera XS (direct drive, 1.75mm)
- **Controller:** BIGTREETECH SKR Mini E3 V3.0 running Marlin firmware

Igus ReBeL 6DOF arm

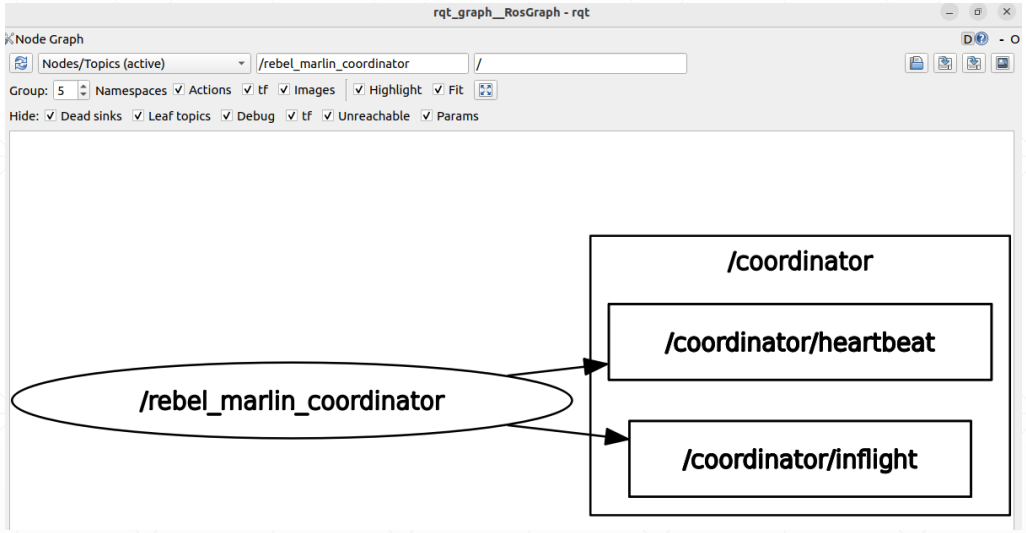


E3D Revo Hemera XS

BIGTREETECH SKR Mini E3 V3.0



System Software Architecture & Communication



System Overview:

- **ROS 2 Node:** /rebel_marlin_coordinator orchestrates the system, runs at 50 Hz
- Publishes key status topics for instrumentation and monitoring:
  - /coordinator/heartbeat: For liveness monitoring.
  - /coordinator/inflight: To track the command queue size in real-time.
- The stack uses ROS 2 Humble with CycloneDDS.

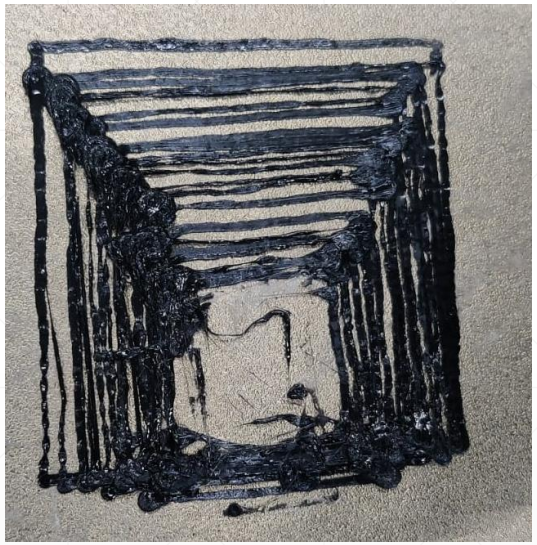
Communication Links:

- **Robot ↔ PC:** Motion commands via TCP/IP using CRI protocol on port 3920. Requires keep-alive every 20–50 ms.
- **Extruder ↔ PC:** G-code commands over USB CDC serial @ 115200 baud. Marlin firmware replies with "ok" acknowledgment.

Experimental Validation Evidence

Objective:

- Bench tests performed at 30 mm/s and 60 mm/s using PETG, 0.4 mm nozzle.
- Deposited line, square, spiral, and lattice coupons on Voron PEI plate.
- Observed bead continuity: No visible starvation or gaps in all trials (n = 6).
- Mean bead width = 0.48 ± 0.03 mm vs. commanded 0.48 mm → < 6 % deviation.



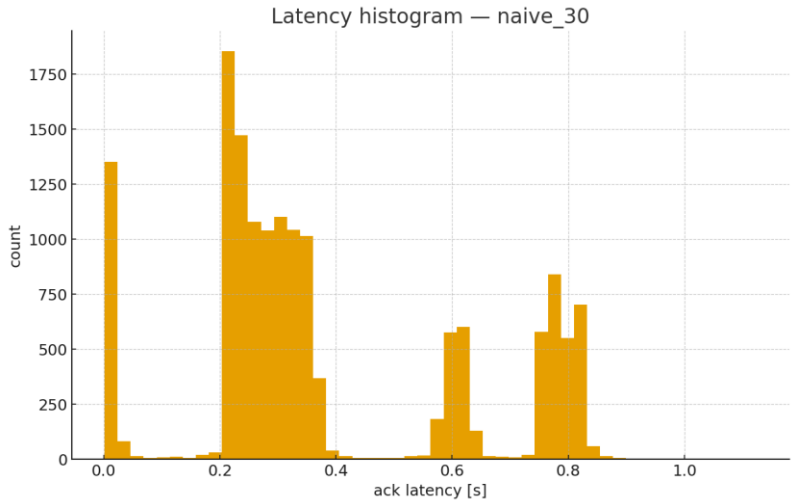
Experimental Validation Print Quality

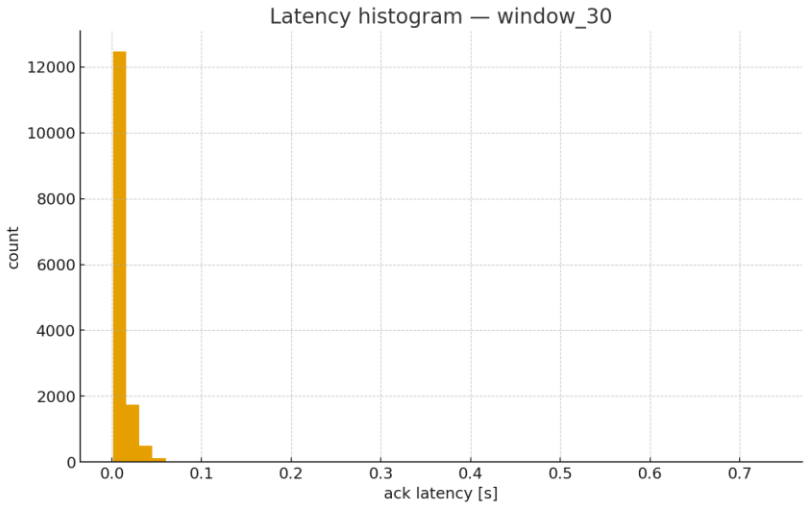
Technical reasons for the defects:

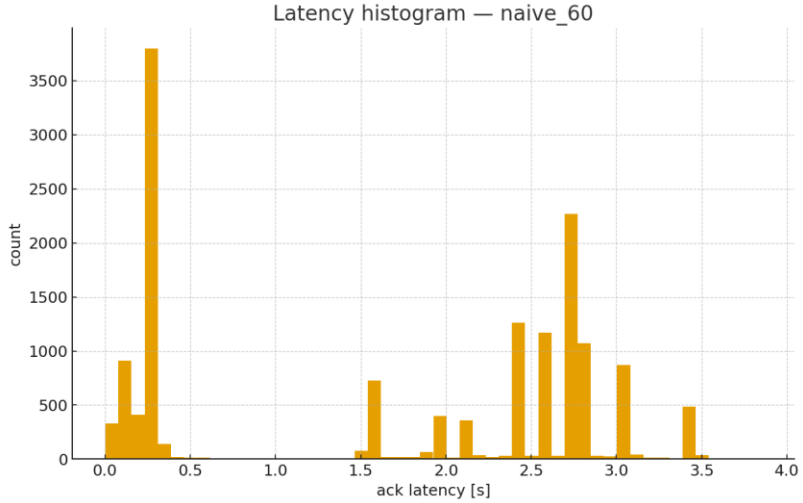
- **Stringing & blobs** = no retraction strategy implemented (extruder keeps oozing).
- **Gaps between lines** = bead width not perfectly tuned to spacing, and minor timing jitter.
- **Uneven deposition** = open-loop extrusion (no closed-loop flow feedback).
- **Layer bonding weak** = only planar paths, not optimized process parameters.
- **Material quality** = PETG is stringy, especially without tuned cooling..

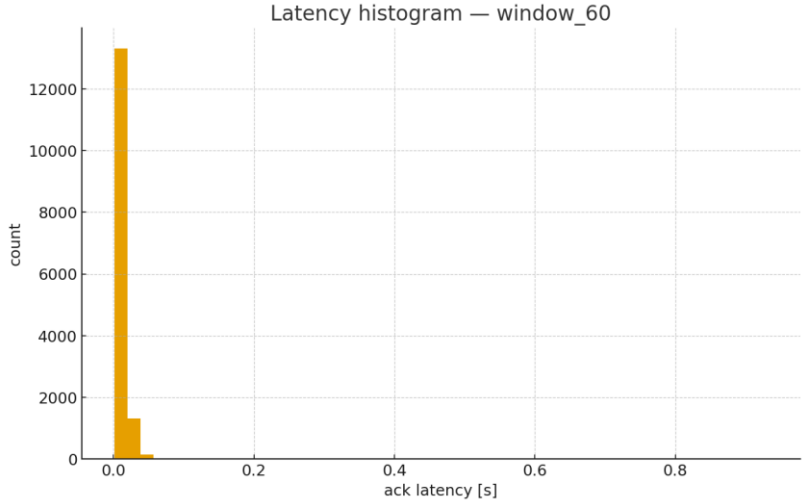
Results

Latency Graphs

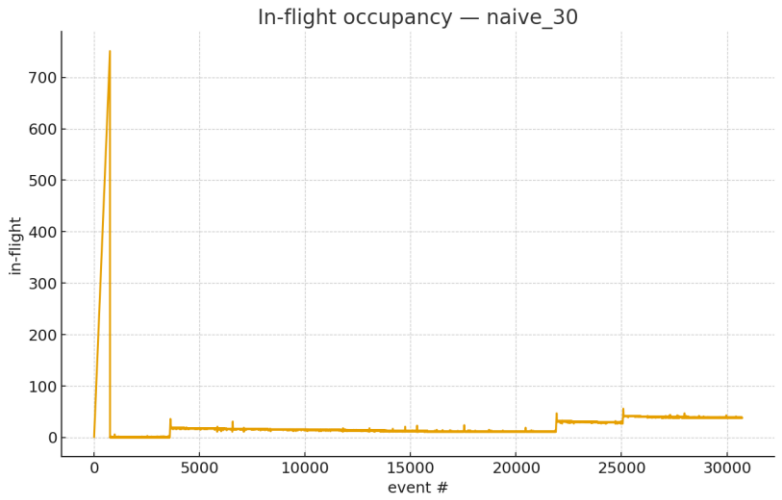


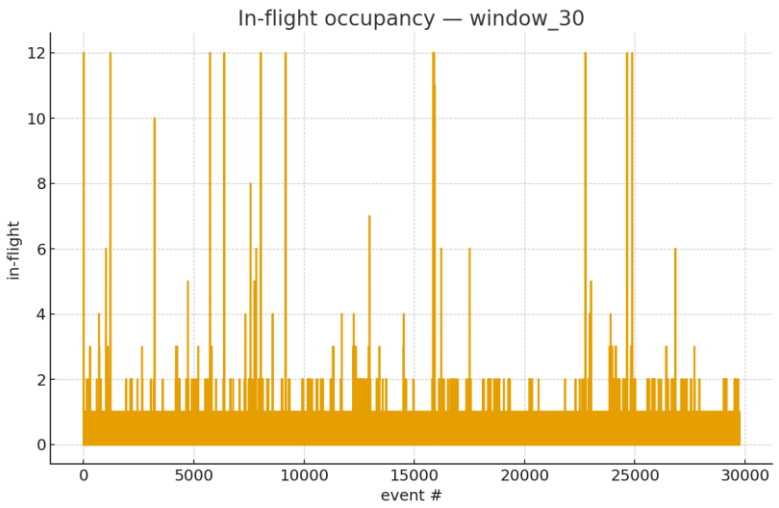


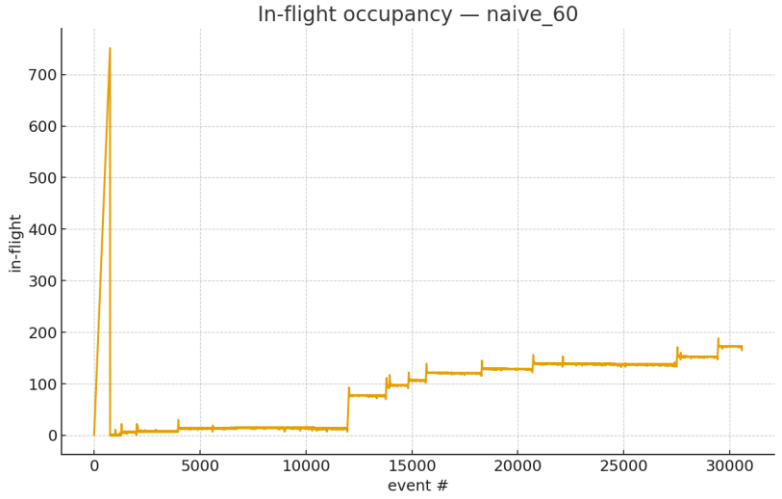


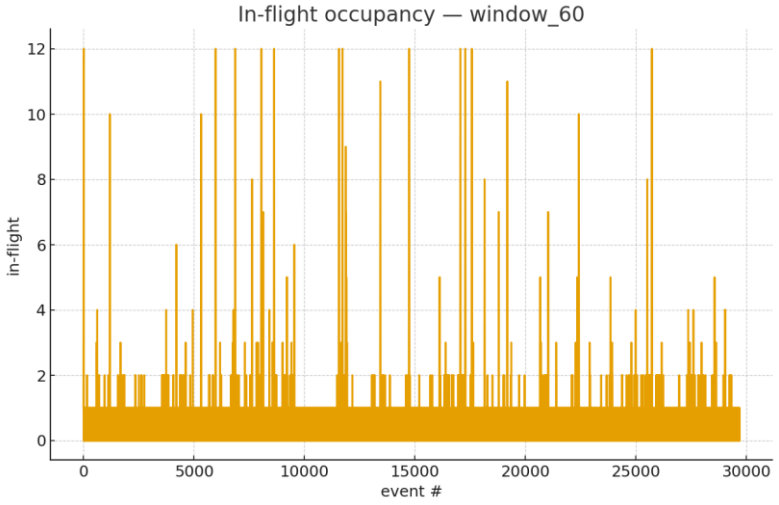


Inflight Graphs









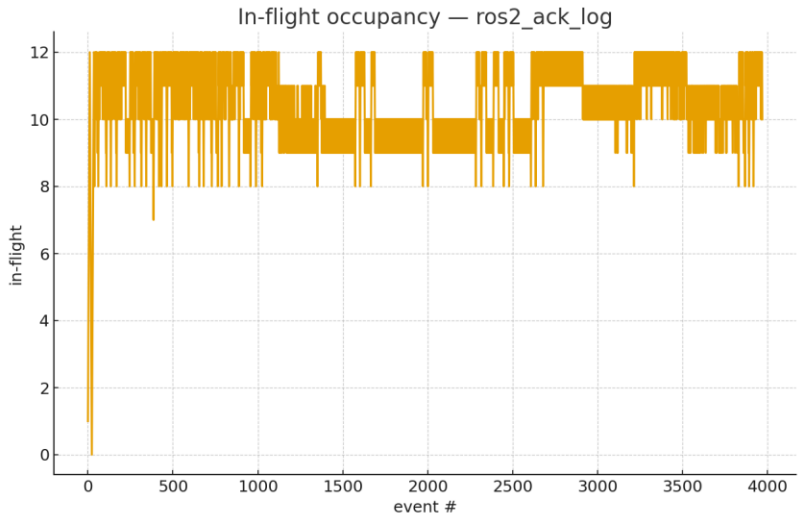
Results

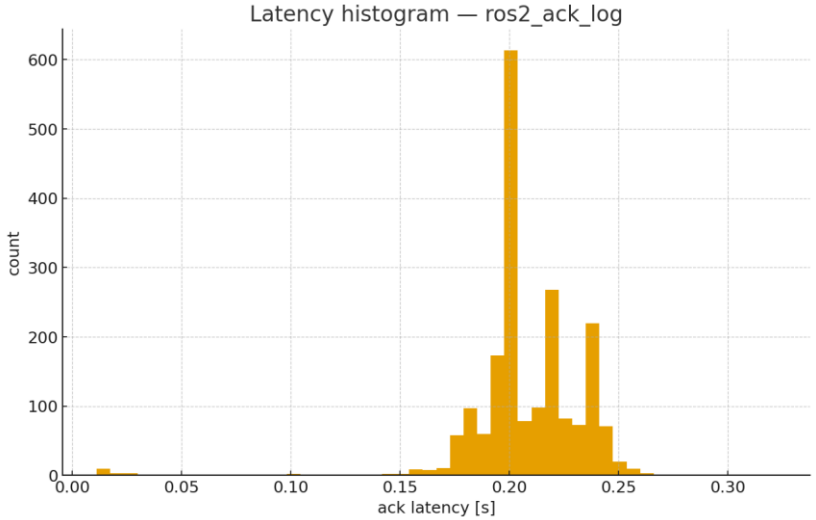
Quantitative Comparison

Metric (at 60 mm/s Equivalent Speed)	Naïve Sender (No Control)	Ack-Aware Window (My Protocol)
Median Acknowledgment Latency (Lok)	2255 ms	3.73 ms
99th Percentile Latency (Tail)	3456 ms	47.1 ms
Planner Backlog (Max In-Flight)	751 Lines	12 Lines (By Design)
Planner Starvation	241 Events	0 Events

Key Takeaway:
The windowing protocol collapses latency from seconds to milliseconds and completely prevents both command backlogs and planner starvation , creating the stable conditions necessary for continuous deposition

Final Validation: Performance in the Full ROS 2 System





In-Flight Occupancy:

- This graph shows the performance of the final ROS 2 node.
- The number of unacknowledged commands is perfectly maintained within the target [8, 12] line window.
- The median occupancy is 10, the 95th percentile is 12, and there are zero starvation events during the run. This proves the windowing logic is working exactly as designed.

Acknowledgment Latency:

- In the full ROS 2 system, the end-to-end service time is higher due to software overhead, with a median latency of ~202 ms and a 95th percentile of ~242 ms.
- This is a critical finding: even though latency is higher than the bench test, it is stable and predictable.

Analysis & Discussion

Discussion

What the Results Mean:

- The core finding is that explicit, host-side pacing is essential for stability on non-deterministic links like USB.
- The window sizing rule, $N_{max} \times \Delta t \gtrsim p_{95}(L_{ok})$ is validated as a robust design principle. The configured 12-line window provides 240 ms of command buffer, which successfully covered the observed p95 latency of ~242 ms in the full ROS 2 system, thus preventing starvation.

Answering the Research Questions:

- Synchronization Settings): A 50 Hz control loop with an [8, 12] in-flight window is proven to be an effective setting for enabling continuous deposition.
- (Robustness): The protocol is inherently robust against the timing jitter present in a standard OS and USB serial link, successfully maintaining a stable command stream under worst-case transport conditions.

Limitations:

- The timing data was captured in extruder-only bench tests to isolate transport-level effects; full motion-coupled prints are future work.
- The system was validated with a single material (PETG) and geometry on planar paths.
- The protocol's performance on deterministic fieldbus transports (e.g., EtherCAT) was not evaluated.

Cost Analysis

Inputs:

- Electricity Price (Germany): e = 0.3943 €/kWh
- PETG Filament Price: F = 19 €/kg
- Total System Power Draw: 231 W
  - Robot (igus ReBeL): 180 W
  - Hotend, Fans & Controller: 51 W
- Bead Geometry: w=0.48 mm, h=0.20 mm

Calculated Operating Costs:

- **Electrical Cost (Constant):**

$$C_{elec}\ [€/h] = \frac{P_{hotend} + P_{fan} + P_{controller} + P_{robot}}{1000} \times e$$

= 0.231 kW × 0.3943 €/kWh = **0.09 €/hour**

Cost at 30 mm/s Speed:

Material cost per hour = $\boldsymbol{C_{mat} = \frac{w \cdot h \cdot v}{1000} \times 3600 \times F \times \rho_{PETG}}$

- Material Cost: 0.25 €/hour
- Materal + Energy Cost: ≈ 0.34 €/hour

Cost at 60 mm/s Speed:

- Material Cost: 0.50 €/hour
- Materal + Energy Cost: ≈ 0.59 €/hour

Cost Analysis

Setup Cost

- Robot arm (igus ReBeL) ≈ 4.970 €
- Extruder + hotend ≈ 149 €
- Controller (SKR + drivers) ≈ 69 €
- Power supply, wiring, fixturing ≈ 34 €

Total setup ≈ 5222 €

$$C_{machine} = \frac{C_{machine\ purchase}}{L_{years} \times H_{annual}} \times t_{print}$$

Depreciation Cost

- Assume 5-year lifetime, 1,000 h/year usage.
- Setup cost ÷ (5,000 h total) ≈ 1.04 €/h depreciation.

Labour Cost:

- Student/researcher/operator ≈ 12.83 €/h
- operator time [h] = R_{labour}

Maintenance Cost:

- Consumables = 100–200 €/year = 0.05 €/h.

Total Cost Would be:

$$C_{print} = C_{material} + C_{energy} + C_{labour} + \frac{C_{machine}}{lifetime\ hours} \times t_{print}$$

At 30 mm/s (material flow ≈ 13.2 g/h)

- Material cost: 0.25 €/h
- Energy cost: 0.09 €/h
- Labour cost: 12.83 €/h
- Machine depreciation: 1.04 €/h
- Overhead (15%): 2.13 €/h

  Total = 16.35 €/h

At 60 mm/s (material flow ≈ 26.4 g/h)

- **Total = 16.63 €/h**

Conclusion

Conclusion

- This thesis successfully designed, implemented, and validated a **practical, host-timed synchronization strategy** that bridges ROS 2 motion control with Marlin-based extrusion on accessible, low-cost hardware.
- The proposed **acknowledgment-aware windowing protocol** is proven to be highly effective. It eliminated planner starvation and command backlog while collapsing acknowledgment latency from seconds to tens of milliseconds on a standard USB link.
- The primary contribution is a **complete, reproducible, and open-source pipeline** that provides a solid foundation for reliable robotic deposition. This work demonstrates that high-performance synchronization is achievable without expensive proprietary hardware or hard real-time operating systems, making advanced robotic AM more accessible.

Future Work

- Integrate full motion-coupled 3-axis toolpaths to validate protocol during dynamic robot motion.
- Evaluate deterministic fieldbus (EtherCAT/CAN) vs. USB for sub-50 ms latency.
- Extend synchronization to curved/conformal layers and multi-material extrusion.
- Implement closed-loop pressure or vision feedback to further improve bead geometry.

References

- **Ding et al., 2016** – Multi-axis robotic AM, large-scale printing.
- **Pan et al., 2019** – Curved layer deposition, structural benefits.
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- https://www.alibaba.com/product-detail/6-axis-industrial-robot-arm-robot_1601229328292.html
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Thank You For The Attention

Raj Manoj Bag